

Cybersecurity challenges in integrated energy systems with cyber-physical integration: A review

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ABSTRACT

Integrated modern energy systems employ advanced information, communication, and control technologies to optimize the coordination of electricity, heat, and gas subsystems, forming interconnected energy cyber-physical systems (IECPS). While this deep integration enhances operational efficiency, it amplifies cybersecurity risks due to the interdependence of cyber and physical components. This paper comprehensively reviews IECPS cybersecurity challenges, focusing on cyber-physical coupling models, security performance assessment, attack methodologies, defense mechanisms, and recovery strategies. The analysis identifies vulnerabilities inherent in coupling models, evaluates security metrics, and classifies attack vectors alongside tailored mitigation approaches. Effective recovery strategies, such as self-healing systems and smart redundancy, are explored to minimize disruptions and bolster resilience. The study underscores critical research gaps, including insufficient multi-timescale modeling and control-level defense frameworks, while advocating for advanced solutions like AI-driven detection, quantum-secure protocols, and blockchain-based integrity verification. By synthesizing cutting-edge research, this work provides a roadmap for developing adaptive, standardized cybersecurity frameworks to safeguard IECPS against evolving threats, ensuring secure and resilient operation in smart energy infrastructures.

Nomenclature

AI	Artificial Intelligence	IECPS	integrated energy cyber-physical systems
ARP	Address Resolution Protocol	IES	integrated energy systems
CHP	Combined Heat and Power	IoT	Internet of Things
CPES	cyber-physical electrical system	MFA	Multi-Factor Authentication
CPS	cyber-physical systems	MITM	Man-in-the-Middle
DNS	Domain Name System	ML	machine learning
DoS	Denial-of-Service	P2G	Power-to-Gas
EG-IES	electricity-gas integrated systems	RES	renewable energy source
EH	Energy hub	SCADA	Supervisory Control and Data Acquisition
EV	electric vehicle	SQL	Structured Query Language
FDI	False Data Injection	SSL	spoofing, Secure Sockets Layer
IDS/IPS	Intrusion Detection/Prevention Systems		

1. Introduction

Modern integrated energy systems (IES) play a pivotal role in advancing sustainable energy development and enhancing operational efficiency [1,2]. The integration of advanced information and communication technologies facilitates optimized system performance and greater penetration of renewable energy sources (RESs) [3,4]. By incorporating renewable generation, electric vehicles (EVs), energy storage systems, and distributed energy resources into the power grid supported by robust communication networks, IES can significantly improve grid efficiency [5]. Additionally, microgrids provide a reliable solution for localized power supply through islanded operation [6]. Driven by global decarbonization efforts, the power sector is rapidly transitioning toward renewable energy, while the transportation sector

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is adopting large-scale vehicle electrification. Sustainable energy targets include integrating 8,000 GW of renewable capacity by 2030, a substantial increase from the current 2,800 GW. Moreover, over 100 countries aim to achieve 100% renewable generation by 2025. Currently, Norway leads with 99% renewable power integration, followed by New Zealand (81%), Brazil (79%), Colombia (74%), Canada (68%), Sweden (67%), and Portugal (65.5%), whereas Saudi Arabia has the lowest adoption rate at 0.1%. Globally, the share of renewable energy in total generation has risen steadily from 26.3% in 2020 to approximately 32.1% in 2024. Given that the transport sector contributes 17% of global CO₂ emissions, the EV market has experienced exponential growth, with over 16.5 million EVs in operation today. Projections indicate that EVs will account for 2% of global electricity demand by 2030 [7,8]. Energy hubs (EHs) are increasingly recognized as a more efficient alternative to conventional grids for integrating RES and EVs [9], as demonstrated in Fig. 1, which illustrates the diverse energy sources and loads within an EH framework.

The concept of EHs offers a way to manage diverse energy systems for a comprehensive sustainable energy model. Integrating technologies like the internet of things, fog computing, and cloud computing enhances EHs, transforming them into smart EHs, but also introduces significant cybersecurity vulnerabilities that must be addressed. Technological innovation is reshaping the energy sector, with demand management playing a crucial role [10]. The growing accessibility of RESs introduces fluctuations in smart EH systems, necessitating flexible strategies to balance supply and demand. smart EH technology and big

data analytics promise more efficient and reliable energy management and greater system resilience. The increasing deployment of smart meters, enabling real-time energy usage communication, and the deregulation of energy delivery have facilitated the smart EH concept, addressing energy demand challenges through two-way communication and energy transfer.

The intermittent nature of RESs necessitates continuous monitoring and real-time communication of load availability and demand data to the controller to ensure efficient grid operation and management. This communication infrastructure is supported by various network models, including the Open System Interconnection model, Transmission Control Protocol/Internet Protocol (TCP/IP), Extensible Authentication Protocol, and specialized microgrid communication protocols [11], as illustrated in Fig. 2. The proliferation of Internet of Things (IoT) devices and architectures has facilitated the deployment of intelligent services across multiple sectors, including smart meters, healthcare, transportation, and the smart grid [12]. Typically, IoT architectures are preferred on the demand side, while the Extensible Authentication Protocol model is implemented on the supply side. The nature of global conflicts is evolving, with cyber warfare emerging as a significant threat. Cyberattacks, which involve unauthorized intrusions into a nation's cyberspace and the disruption of critical communication channels, are increasingly being recognized as a future warfare strategy [13]. Even the most technologically advanced nations remain vulnerable to such threats. As a result, governments worldwide are prioritizing cybersecurity awareness and infrastructure development [14]. According to the

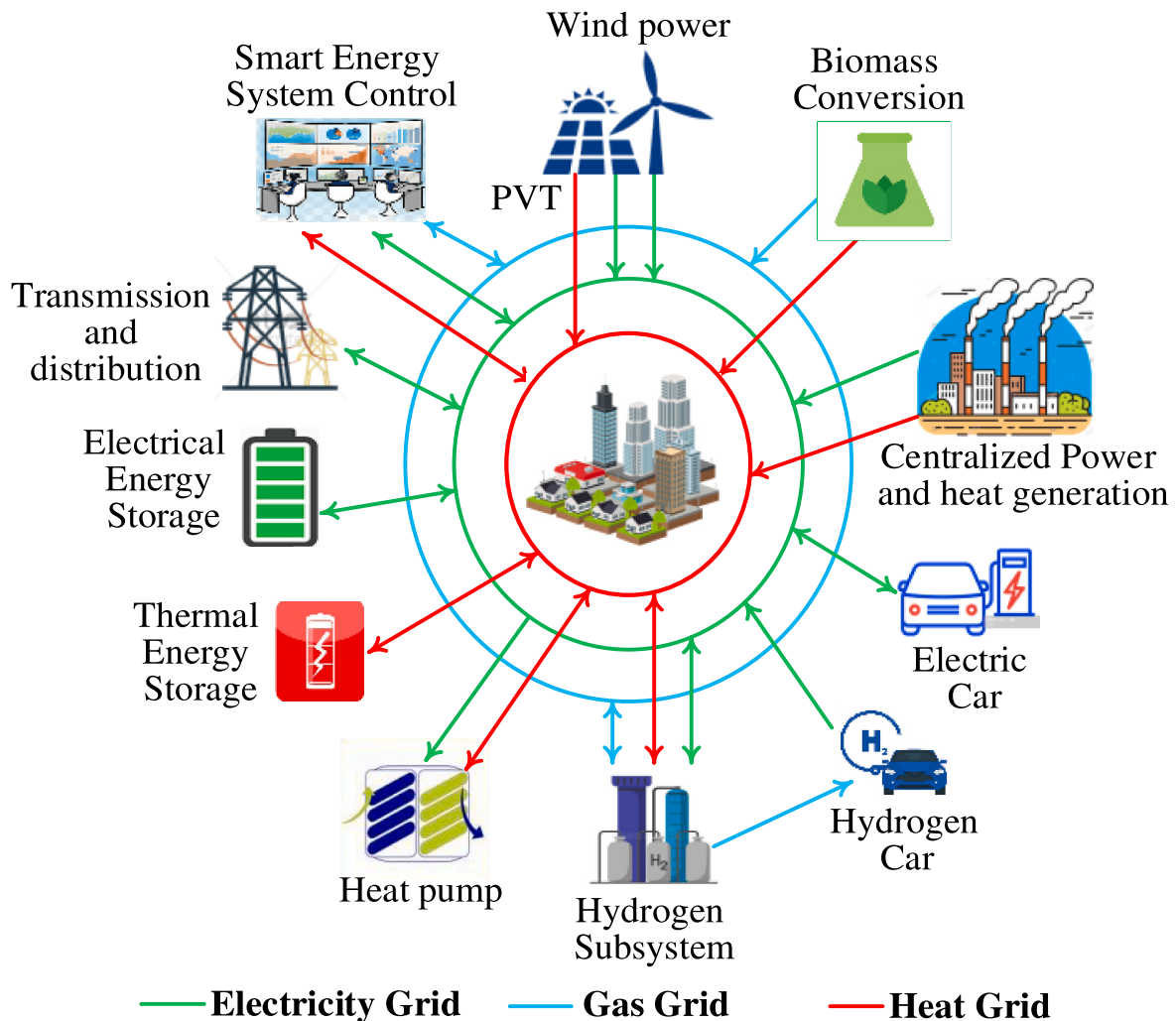


Fig. 1. Overview of a multi-energy system.

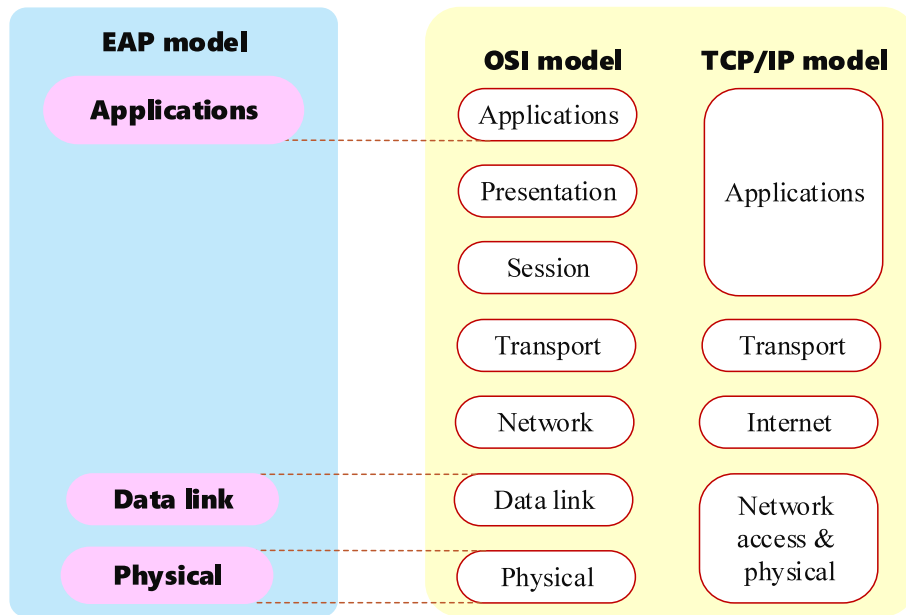


Fig. 2. Communication models for multi-energy carrier system.

Crunchbase Cybersecurity Report [15], cybersecurity funding has surged by nearly 700% over the past decade. The United States accounts for the majority of global cybersecurity investments (76%), followed by Israel (13%) and the United Kingdom (3%), while all other countries collectively represent 8%. Cyber adversaries employ a range of attack strategies to compromise system security, with seven primary attack types identified, as illustrated in Fig. 3.

The evolution of integrated energy cyber-physical systems (IECPS)

has significantly enhanced the performance and efficiency of IES [16]. However, the convergence of diverse information technologies and energy infrastructures has increased system complexity, introduced new vulnerabilities [17], and heightened risks to operational security particularly in the realm of cybersecurity. These risks include communication delays, system failures, and the growing threat of malicious cyberattacks [18]. The landscape of cybersecurity in energy systems has undergone significant conceptual shifts. Initially, research focused on

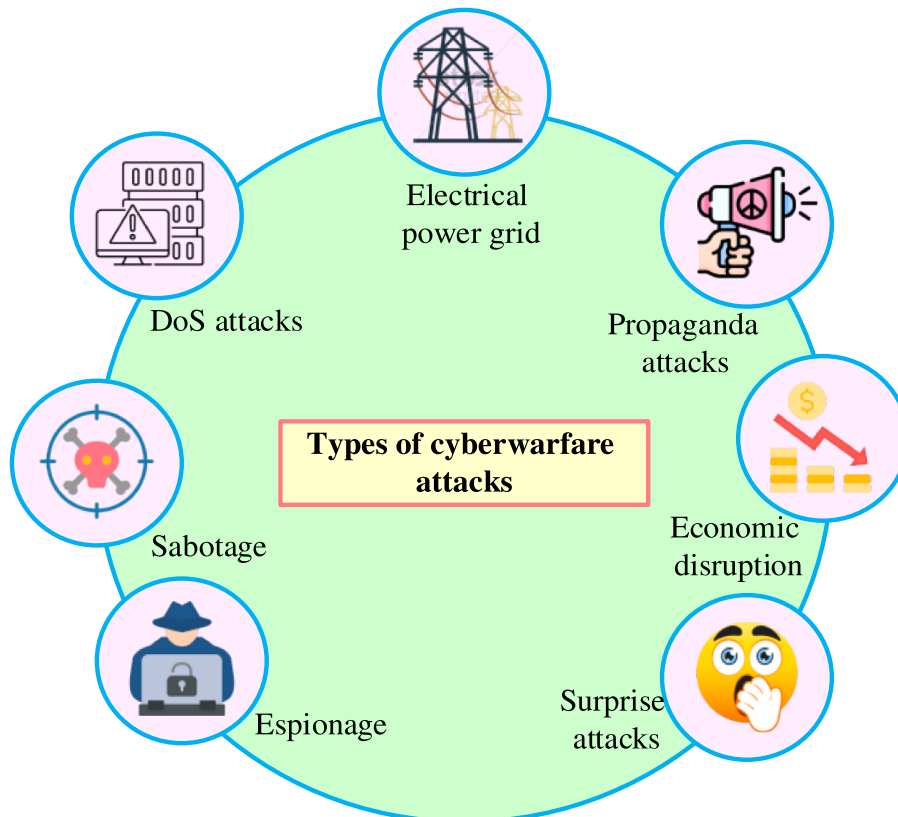


Fig. 3. Different attack methodologies used for cyber-warfare.

isolated IT/OT security within traditional power networks. With the advent of smart grids, the focus shifted toward cyber-physical security to address DER-rich environments. Today, the field is evolving toward fully integrated, multi-carrier IECPS, characterized by deep cross-domain dependencies between electricity, gas, and thermal networks. This evolution reflects a transition from securing individual components to ensuring the resilience of interconnected infrastructures.

Recent cyber-related disruptions in energy systems underscore these vulnerabilities. Notable incidents include the 2015 Ukraine power grid cyberattack [19] and the 2021 Colonial Pipeline attack [11], alongside numerous operational outages caused by software application failures and communication disruptions [20]. While extensive research has been conducted on various aspects of IES, including planning, scheduling, simulation, and market analysis [21,22], cybersecurity remains a critical challenge [20]. Traditional physical security measures [23] are inadequate in addressing the unique threats posed by cyber-physical integration. As highlighted in Ref. [24], the implementation of large-scale cyber-physical systems (CPS) within IES presents significant challenges, particularly in cybersecurity. To ensure the stable operation of IES, research on IECPS cyber-physical integration has expanded, focusing on areas such as coupling modelling, security performance assessment, attack and defence mechanisms, and fault recovery strategies. Although cybersecurity reviews exist for power systems [25], a comprehensive and systematic review of cybersecurity challenges specific to IECPS remains an unexplored area in the literature.

As summarized in Table 1, while existing literature has extensively covered cybersecurity in isolated domains, a significant research gap remains. For instance, the review by Du et al. in Ref. [26] focused primarily on electrical power systems, and Li et al. restricted its scope to building automation in Ref. [27]. Although recent works in Ref. [28] provided comprehensive attack impact analyses, they do not account for the complex interdependencies and vulnerability propagation across multiple energy carriers (electricity, gas, and heat) characteristic of IECPS.

While extensive research has been conducted on cybersecurity in energy systems, the majority of these studies have primarily focused on traditional power systems or electrical power grids, often neglecting the broader security challenges posed by IECPSs. Unlike conventional power systems, IECPSs involve the interconnection of multiple energy carriers, including electricity, thermal, gas, etc. within a unified operational framework. This intricate integration increases system complexity and introduces new cybersecurity vulnerabilities, such as cross-domain cyber threats, cascading failures, and multi-vector attack surfaces, which have not been adequately explored in existing literature. Furthermore, conventional cybersecurity approaches tailored for electrical power grids may not be directly applicable to IECPSs due to their heterogeneous structure, dynamic interactions, and diverse communication protocols. Addressing these unique security challenges requires a holistic, interdisciplinary approach that considers the interdependencies between cyber and physical components across different energy domains. This research aims to bridge this critical gap by:

- Analyzing the impact of cyber-physical system integration on security risks and developing a model to illustrate its influence on overall system security.
- Providing a comprehensive review of cybersecurity attacks on integrated systems, classifying them based on vulnerabilities and their operational consequences.
- Proposing an artificial intelligence (AI)/machine learning (ML)-based detection approach for enabling faster and more accurate identification of sophisticated attacks such as False Data Injection (FDI) and Man in the Middle (MITM) attacks.
- Suggesting post-attack recovery strategies by incorporating Self-Healing System concepts to enhance resilience and emergency response.
- Conducting an analytical case study of the 2015 Ukrainian power grid attack, showing how modern cybersecurity measures could assist in preventing or mitigating such incidents.

This research presents an integrated framework that systematically addresses the technical, offensive, and defensive aspects of cybersecurity in integrated energy systems. Specifically, it identifies potential attack vectors and system vulnerabilities, evaluates the technical architecture and communication infrastructure prone to cyber threats, and develops detection, prevention, and response mechanisms using advanced technologies such as ML and self-healing systems.

The remaining structure of the paper is organized as follows: Section 2 outlines the research methodology, adhering to PRISMA guidelines to ensure a systematic and transparent literature review. Section 3 details the structural framework of IECPS, examining the physical and cyber layers as well as the coupling characteristics between subsystems such as electricity, gas, and thermal. Section 4 describes modeling approaches for IECPS, covering physical layer dynamics, cyber-physical electrical system (CPES) interactions, and fully integrated multi-energy frameworks. Section 5 presents an in-depth analysis of cyberattack methodologies, specifically FDI, MITM, and Denial of Service (DoS). Section 6 evaluates the operational safety of IECPS through comprehensive reliability and vulnerability assessments, emphasizing the impacts of cyber-physical coupling. Section 7 explores cyberattack mitigation and security architecture, categorizing strategies into resource optimization, intelligent detection, and collaborative scheduling. Section 8 discusses cybersecurity-oriented resilience and recovery frameworks, including advanced mechanisms like smart redundancy and self-healing systems. Section 9 provides a detailed case study of the 2015 Ukraine power grid attack, analyzing the incident history and potential mitigation through modern AI/ML methods. Section 10 outlines future challenges, such as quantum-secure cryptography and blockchain integration. Finally, Section 11 concludes the study by highlighting research gaps and proposing directions for advancing IECPS cybersecurity frameworks.

2. Research methodology

To ensure the replicability of this review, a systematic approach following the PRISMA guidelines was adopted. The literature search was conducted across major databases, including IEEE Xplore, ScienceDirect, Scopus, and Web of Science, covering the period from 2015 to 2025. The search strings utilized keywords such as 'Integrated Energy Systems', 'Cybersecurity', 'Cyber-Physical Systems', and 'multi-carrier networks'. Studies were selected based on their relevance to cross-sectoral coupling and cybersecurity impacts, while purely electrical or purely thermal papers without cyber-physical integration were excluded. This review, as shown in Fig. 4, follows a transparent and structured methodology inspired by PRISMA guidelines to enhance reproducibility and transparency.

2.1. Databases and sources

The literature search was conducted using IEEE Xplore, Scopus, Web

Table 1
Comparison of the present review with existing representative surveys.

Feature	[26]	[27]	[28]	This Review
Main Scope	Power Systems	Building Automation	Power Systems	IECPS
Energy Carriers	Electricity only	HVAC/Power	Electricity only	Power, Gas, Heat, H ₂
Coupling Models	Limited	None	None	In-depth CP Coupling
Cross-Carrier Attacks	No	No	No	Yes (Detailed Analysis)
Recovery Strategies	Basic	No	No	Coordinated Resilience

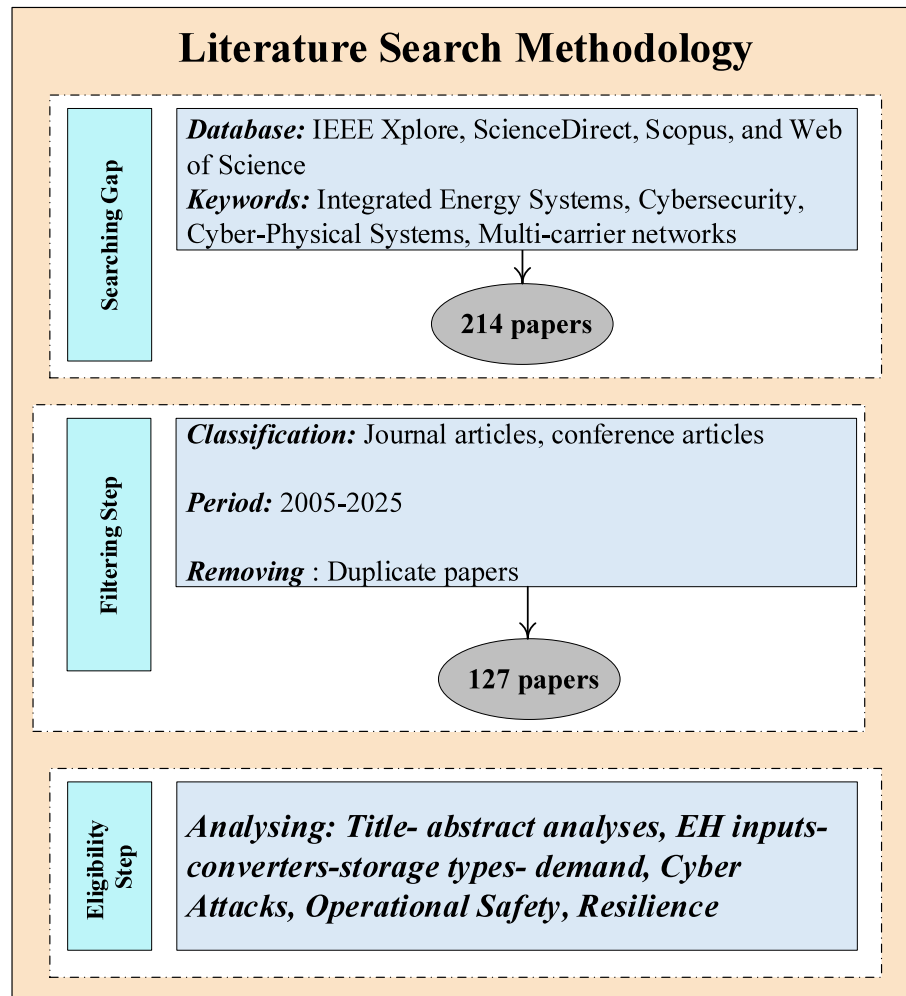


Fig. 4. The methodological principle for previous research and review applied in this paper.

of Science, and ScienceDirect to ensure comprehensive coverage of interdisciplinary research on integrated energy cyber-physical systems (IECPS).

2.2. Search strategy

The search used combinations of keywords including: ("Integrated Energy Systems" OR "Multi-energy systems" OR "Energy Internet" OR "Integrated Energy Cyber-Physical Systems") AND ("Cybersecurity" OR "Cyber-physical security" OR "Cyber attacks" OR "Intrusion detection" OR "Resilience").

2.3. Time span

Publications from 2015 to 2025 were considered to capture both foundational and recent developments.

2.4. Inclusion and exclusion criteria

Studies were included if they were peer-reviewed, addressed cyber-security or resilience in cyber-physical energy systems, and considered system-level or multi-carrier impacts. Studies focusing solely on traditional IT security or isolated components without physical relevance were excluded.

2.5. Screening and selection process

Titles and abstracts were initially screened, followed by full-text assessment of eligible studies. A PRISMA-style flow diagram summarizes the identification, screening, eligibility, and inclusion stages.

2.6. Evaluation metrics, resilience quantification, and decision relevance

Due to the heterogeneity of IECPS architectures, energy carriers, threat models, and evaluation setups, a quantitative meta-analysis is not feasible. Instead, a structured qualitative synthesis is adopted to unify decision-relevant metrics across multiple dimensions. Cybersecurity performance is evaluated using security-oriented metrics such as detection coverage, exploitability, false alarm rate, mean time to detect, and mean time to respond. Reliability and resilience metrics capture physical-system impacts, including expected energy not supplied, cyber-extended SAIDI/SAIFI, survivability, and post-attack recovery trajectories. Economic metrics quantify operational and market impacts through imbalance costs, demand response penalties, and value of lost load (VoLL). For multi-carrier IECPS involving thermal, gas, or hydrogen infrastructures, safety-oriented metrics such as over-temperature, over-pressure, and constraint violation durations are also considered.

2.7. Mitigation strategy synthesis

Mitigation approaches are synthesized by analyzing trade-offs among latency, determinism, deployment cost, and operational

feasibility. While AI-based detection techniques often achieve high accuracy, their deployment in OT environments may be constrained by real-time requirements and certification barriers. In contrast, physics-informed and rule-based methods provide higher determinism and interpretability, making them suitable for safety-critical IECPS applications.

2.8. Standards, regulation, and governance mapping

To support practical deployment, this review maps major cybersecurity standards and regulations (e.g., NERC CIP, IEC 62443, IEC 62351, ISO/IEC 27019, and OCPP security) to IECPS layers and energy carriers. Identified gaps include limited coverage of DER aggregators, behind-the-meter assets, and thermal and hydrogen interfaces, highlighting the need for integrated compliance roadmaps. Although no statistical meta-analysis is performed, the unified metric taxonomy, mitigation trade-off synthesis, and standards mapping provide a structured, transparent, and decision-relevant methodological foundation for this review.

2.9. PRISMA-style

This review follows a transparent and reproducible PRISMA-style methodology. Literature was collected from IEEE Xplore, Scopus, and Web of Science using structured search strings related to cybersecurity, integrated energy systems, and cyber-physical systems. Studies published between 2015 and 2025 were considered. Inclusion criteria comprised peer-reviewed journal articles and conference papers focusing on cyber-physical security in energy systems. Exclusion criteria included non-English publications, purely economic studies without cyber-physical relevance, and papers lacking technical validation.

The selection process involved identification, screening, eligibility assessment, and final inclusion stages, summarized in a PRISMA flow diagram. [Table 2](#) Mapping of major cybersecurity standards, regulations, and governance frameworks to system layers and energy carriers in integrated energy cyber-physical systems, highlighting applicability and existing gaps.

3. Structural framework of integrated energy cyber-physical systems

IES, combining diverse energy sources, is further enhanced by integrating CPS, creating IECPS. While IECPS offer operational and control advantages, their complexity due to the tight cyber-physical coupling introduces significant cybersecurity vulnerabilities. Therefore, a deep understanding of the IECPS structure including the interaction of cyber

and physical components, communication pathways, and control systems is crucial. This analysis examines these structural intricacies to identify attack vectors, understand threat propagation, and develop robust cybersecurity strategies for ensuring the resilience of these critical infrastructures. The analysis will cover key elements like energy subsystems, communication networks, control centres, and data units, providing a comprehensive overview of their interactions.

3.1. Physical and cyber structure of IECPS

The IECPS structure, as depicted in [Fig. 5](#), comprises a complex multi-energy flow physical system, encompassing electrical, gas, thermal, cooling, and hydrogen systems, with diverse equipment including sources, networks, loads, energy storage, and multi-energy coupling equipment. At the cyber level, the development varies across subsystems; the electrical system's cyber infrastructure is relatively advanced, while gas, thermal, cooling, and hydrogen systems lag [\[24\]](#). Furthermore, cyber systems between sub-energy systems are largely independent due to confidentiality surrounding detailed system structures and equipment parameters, limiting information exchange primarily to scheduling and trading.

3.2. Physical and cyber characteristics of IECPS

[Table 3](#) presents a comprehensive classification of coupling devices according to their roles as energy sources or loads within five key subsystems of an IES: Gas, Electrical, Thermal, Cooling, and Hydrogen. It highlights the complex interdependencies and bidirectional energy exchanges characteristic of cyber physical multi energy systems. Illustrative examples include the cogenerative function of Combined Heat and Power (CHP) units, and the dual conversion capabilities of Power to Gas (P2G) and Fuel Cells between electrical and hydrogen systems. Technologies such as Heat Pumps and Absorption Chillers reflect the integration of electrification into thermal and cooling domains, supporting decarbonization goals. Additionally, adaptable components like Thermal Energy Storage enhance system flexibility and energy balancing. Overall, the table captures the multifaceted energy flows and technological synergies fundamental to the design and operation of sustainable, integrated energy infrastructures.

The cyber layer of current IECPS sub energy systems is relatively independent, featuring separate secondary equipment, SCADA systems, with information coupling limited to energy supply/demand and market transactions; thus, sub energy systems are independent CPSs, while IECPS is a higher-level CPS. Within a sub energy CPS like the CPES, the physical layer comprises devices like generators and transformers, while the cyber layer includes functions like state estimation and voltage

Table 2
Mapping of cybersecurity standards, regulations, and governance frameworks in IECPS.

Standard/Regulation	System Layer	Energy Carrier	Scope/Coverage	Limitations & Gaps in IECPS
NERC CIP	Transmission, Control Centres	Electricity	Critical infrastructure protection, access control, incident reporting	Limited applicability to distribution systems, DERs, aggregators, and behind-the-meter assets
IEC 62443	Device, Control, Network	Electricity, multi-energy	Industrial automation and control system security, defense-in-depth	Not explicitly designed for multi-carrier energy coupling or market-layer interactions
IEC 62351	Communication & Control	Electricity	Securing power system communication protocols (e.g., IEC 61850, Supervisory Control and Data Acquisition (SCADA))	Limited coverage of IoT-based DERs, cloud platforms, and cross-sector energy systems
ISO/IEC 27019	Organizational & IT/OT	Electricity	Information security management for energy utilities	Focuses mainly on electrical utilities; limited guidance for thermal, gas, and hydrogen systems
OCPP Security Extensions	Device & Backend	EV Charging	Authentication, encryption, and secure communication for EV charging infrastructure	Narrow scope limited to EV ecosystems; weak integration with grid-wide IECPS security
OEM Grid Codes & Vendor Policies	Device & Plant	Electricity, Renewables	Equipment-level security and operational constraints	Lack of standardization; vendor-specific and often opaque
Data Protection Regulations (e.g., GDPR)	Market & Cloud	Multi-energy	Data privacy, consumer protection, data governance	Do not directly address cyber-physical risks or real-time operational security

Integrated Energy Systems

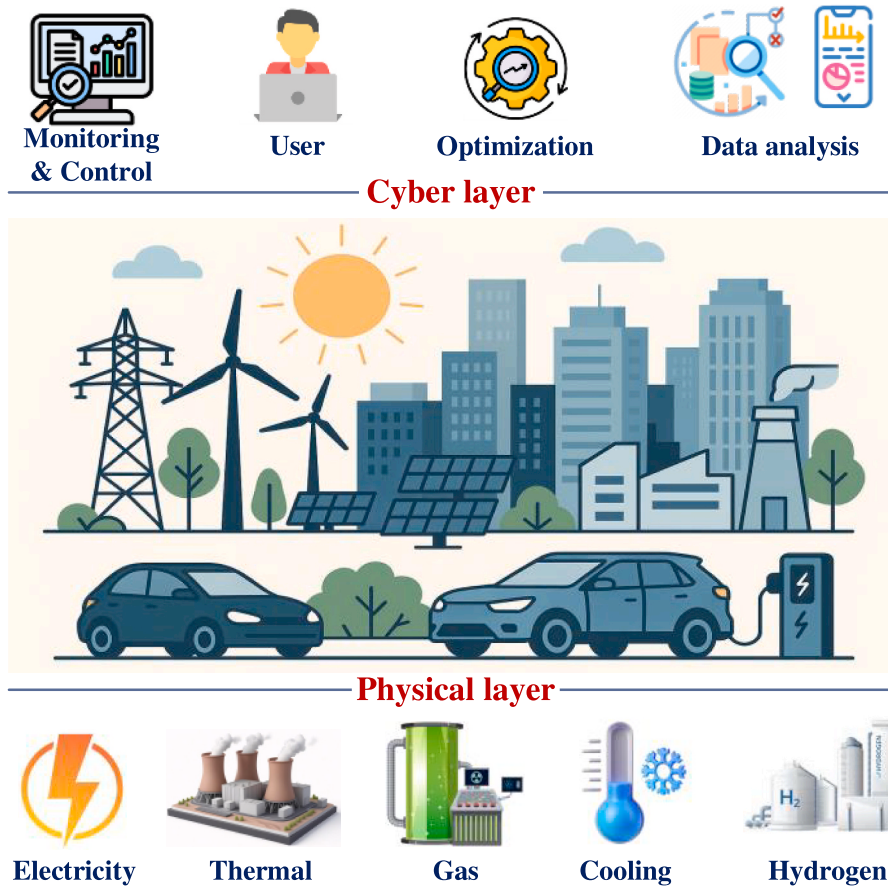


Fig. 5. An illustrative diagram of the IECPS structure.

Table 3
Roles of energy conversion technologies in multi-energy systems.

Coupling Device	Gas System	Electrical System	Thermal System	Cooling System	Hydrogen System
CHP Unit	Load	Source	Source	–	–
Electric Boiler	–	Load	Source	–	–
Gas Boiler	Load	–	Source	–	–
Electrolyzer (H ₂ G)	–	Load	–	–	Source
Heat Pump	–	Load	Source	Source	–
Absorption Chiller	–	–	Load	Source	–
Fuel Cell	–	Source	Source	–	Load
Power-to-Gas (P ₂ G)	Source	Load	–	–	Load
Thermal Energy Storage	–	–	Source/Load*	–	–
Gas Turbine	Load	Source	–	–	–
Electric Chiller/AC	–	Load	–	Source	–

Notes: “Source”: Device supplies energy to the subsystem; “Load”: Device consumes energy from the subsystem; “–”: No interaction with the subsystem; Reversible devices may operate in dual modes (e.g., heat pumps for heating/cooling); and, Storage devices act as loads when charging and sources when discharging.

control; secondary equipment and communication networks form the cyber-physical coupling layer, where operational information is collected, processed, and transmitted to cyber system for control

decisions [25].

IECPS, compared to sub-energy CPSs, exhibits a higher degree of cyber-physical coupling and interdependence due to increased information interaction and coupling devices (e.g., CHP units), intensifying physical interdependence and creating logical dependencies between electrical, thermal, and gas systems, complicating cybersecurity analysis. Therefore, comprehensive IECPS cybersecurity evaluation requires considering both sub-energy system coupling and the complex interdependencies across all layers, necessitating effective strategies that account for these interactions to ensure safe and reliable operation.

4. Modeling approaches for IECPS

Developing effective cybersecurity strategies for IECPS requires accurate and comprehensive modelling approaches that capture the multi-domain physical dynamics and their interaction with cyber layers. The modelling of IECPS can be broadly classified into three interrelated layers: (1) physical layer modelling of multi-energy systems, (2) CPES, and (3) fully integrated cyber-physical modelling of IECPS. Each layer reflects a different degree of integration and complexity, as well as distinct challenges and methodologies.

4.1. Physical layer modelling

IECPS physical modelling strategies generally diverge into two distinct paths, as contrasted in Table 4. The first approach, Separate Modelling, involves separately modelling individual energy systems (e.g., electricity, heat, gas) and then coupling them based on energy flow

Table 4

Comparison between separate and unified modelling approaches for integrated energy systems.

Comparison Aspect	First Approach: Separate Modelling	Second Approach: Unified IES Modelling
Method	Model each energy system (electricity, heat, gas) separately, then couple them	Develop a unified model that includes all systems together
Coupling Technique	Based on energy flow relationships between systems	Unified models such as the “Energy Hub” or unified time-domain models
Challenges/ Limitations	Difficulty handling complex multi-energy flows and differing time scales	More complex to develop initially, but overcomes coupling and integration issues
Capability for Integrated Representation	Limited, especially with complex interactions	Comprehensive and better suited for analysing integrated energy systems

relationships. A critical advantage of this approach is the ability to use established domain-specific simulators. However, as noted in the literature, this method fundamentally struggles with complex multi-energy flow couplings and differing time scales. Since electrical transients differ significantly from thermal or gas dynamics, separate models often fail to capture the rapid destabilization dynamics required for analyzing cyberattacks [26–28].

In contrast, the Unified Modelling approach offers a more comprehensive framework. By employing unified time-domain models or circuit theory extensions, this approach overcomes integration issues. However, the trade-off is often computational complexity; while comprehensive, these models face challenges regarding the high computational cost of real-time simulation, limiting their utility for large-scale, real-time security assessments compared to simpler separate models [29,30].

4.2. Modelling of CPES

Modelling the CPES requires capturing the interaction between electrical infrastructure and cyber components. Current methodologies generally fall into two categories with distinct trade-offs. Information Interaction Models focus on the relationship between cyber and physical systems, where information functions reflect module dynamics and state awareness to achieve cyber-physical model matching. While robust for analyzing data exchange, these models often focus primarily on observability [31]. Conversely, Coupling-Association Models, utilizing methods such as Petri nets [32,33], graph models, and complex networks [34,35], excel at verifying logical dependencies. For instance, finite random Petri nets can create a unified cyber-physical model. A critical limitation, however, is that these discrete models often struggle to represent the continuous physical dynamics necessary for stability analysis. Consequently, recent literature is shifting toward Hybrid Dynamic Models that can simultaneously capture both continuous physical dynamics and discrete cyber events, offering the fidelity required for modern security assessments that pure Petri nets or graph models lack [36,37].

4.3. Modelling of IECPS

While CPES models focus on electricity-based systems, IECPS cyber-physical modelling aims to extend these methodologies to fully integrated, multi-energy systems. This includes incorporating cyber components not only for electricity, but also for thermal and gas networks, and capturing their coordinated operation under cyber-physical constraints [38,39].

Despite these advancements, a critical gap remains in the fidelity of current IECPS models. Currently, most models use simplified logical

coupling between energy and information layers, with limited consideration for system-wide control strategies or advanced communication protocols [40–42]. Consequently, these models frequently fall short in accurately reflecting the complexity of modern infrastructures, particularly when subjected to sophisticated cyber-attack scenarios. The primary barriers to more robust modeling include the heterogeneity of involved systems, the lack of standardized interfaces for cyber-energy integration, and the high computational cost associated with real-time simulation. Future research must therefore prioritize integrated cyber-physical frameworks utilizing approaches such as co-simulation (combining electrical simulators with network emulators [43]) and hybrid dynamic models that capture both continuous physical dynamics and discrete cyber events [44].

A deeper understanding of the modelling landscape across physical, CPES, and IECPS levels highlights the evolution from component-level representations to integrated, cyber-aware frameworks. Table 5 summarizes the key features, advantages, and limitations of each approach.

5. Cyber attacks in cyber-physical systems: threats and real-world cases

CPSs are inherently susceptible to a variety of cyberattacks due to their nature, which combines computational and physical components. Among the most prevalent, impactful, and frequently exploited attack vectors are FDI attacks, MITM attacks, and DoS attacks. These attack types are particularly favoured by malicious actors for their ability to compromise system integrity, disrupt operations, and exploit communication vulnerabilities within CPS architectures.

5.1. False Data Injection attacks

FDI attacks represent the most pervasive cyber threat to

Table 5

Comparative analysis of modelling approaches for IECPS.

Aspect	Physical Layer Modelling	CPES Modelling	IECPS Cyber-Physical Modelling
Scope	Focused on physical energy flows (electricity, gas, heat)	Electrical systems + cyber infrastructure	Multi-energy systems + cyber + control layers
Systems Covered	Electricity, heat, gas (individually or coupled)	Electricity-dominant	Electricity, heat, gas, storage, control.
Cyber Integration	None or implicit	Partial (power-focused)	Full (across energy domains)
Modelling Tools	Energy Hub, Equivalent Circuits, Time-Domain Simulators	Petri Nets, Graph Models, State Transition Models	Co-simulation, Hybrid Dynamic Systems, Formal Logic
Temporal Resolution	Continuous (physical dynamics)	Mixed (cyber discrete + physical continuous)	Hybrid (cyber events + multi-timescale physics)
Interdependency Representation	Limited (via energy flow equations)	Moderate (information exchange)	High (control, communication, cyber threats)
Use Cases	Energy planning, optimization, design	Cybersecurity, stability analysis, anomaly detection	Security assessment, integrated control, cyber-resilience
Strengths	Clear physical laws, easier simulation	Captures attack surfaces, cyber-physical interactions	Realistic for complex, real-world systems
Limitations	Ignores cyber dynamics, limited realism	Often domain-specific, simplified control logic	High complexity, lack of standard frameworks, heavy data needs

Table 6
Mapping cyber-attacks to physical impacts in IES.

Attack Type	Targeted IES Component	Primary Physical Impact	Cross-Carrier Secondary Impact
FDI Attack	Gas Turbine Controller	Gas pressure fluctuations	Loss of electrical generation/Voltage drop
DoS Attack	EH Central Controller	Delayed scheduling commands	Energy imbalances & increased operational costs
MITM Attack	Smart Meters/Demand Response	Unauthorized load shedding	Disruptions in thermal comfort & gas demand

decentralized infrastructures, particularly within microgrids and IES. Unlike traditional systems, FDIAs in IES exploit the cyber-physical coupling to bypass bad data detection by crafting stealthy vectors that align with the system's physics [45]. While attackers may target physical sensors, compromising communication links such as those utilizing Modbus or IEC 61850 protocols is more common due to inherent lack of encryption in legacy OT environments [46].

The primary objective is to manipulate sensor readings (e.g., bus voltages, gas flow rates, or heat temperatures), which can mislead the energy management system into making sub-optimal dispatch decisions. This leads not only to financial losses but also to critical 'cross-carrier' imbalances; for instance, a manipulated gas pressure reading can cause an incorrect ramp-up of a CHP unit, destabilizing the connected electrical feeder [47]. Beyond web-based vulnerabilities like SQL injection or cross-site scripting often found in monitoring interfaces, advanced FDIAs utilize topology-aware algorithms to remain undetectable while inducing physical violations like line overloading or frequency deviations [48]. An example of how these attacks distort sensor readings and impact the physical equilibrium is illustrated in Fig. 6 [11].

FDI attacks can be categorized based on the attacker's level of system knowledge and access. If the attacker has complete system knowledge and access to critical infrastructure, the attack is considered internal. These internal attacks are the most dangerous and effective because the attacker's understanding of the system allows them to craft sophisticated and difficult-to-detect attacks. External attacks, on the other hand, are carried out by those with limited system knowledge, relying instead on exploiting weaknesses in the communication network. Additionally, based on the attacker's knowledge and motives, these attacks can be further classified into various types, such as continuous, intermittent, stealthy, constrained, unconstrained, and time-varying.

Let us consider $\mathbf{A}_i$ the input vector with n samples, $\mathbf{A}_i = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \dots, \mathbf{a}_n]$. The attacker injects the false data $\mathbf{F}_i$ into the input vector, where $\mathbf{F}_i = [\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \dots, \mathbf{f}_n]$. Equation (1) represents the modified input vector resulting from an FDI attack, where malicious data has been deliberately introduced to compromise the integrity of the system's measurements or control signals.

$$\mathbf{C}_i = \mathbf{A}_i + \mathbf{F}_i \quad (1)$$

An attack is considered a continuous FDI attack when the attack continues till the end of the vector from its start. For instance, if the attack is initiated at the third sample, it propagates until the n th sample. Equation (2) denotes input vector after a continuous FDI attack.

$$\mathbf{C}_{cont} = [\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3, \dots, \mathbf{c}_n] \quad (2)$$

where $\mathbf{C}_{cont}$ denotes the input vector, $\mathbf{c}_1 = \mathbf{a}_1$ and $\mathbf{c}_2 = \mathbf{a}_2$, and other samples are attacked samples. In a continuous FDI attack, $\mathbf{attack}_{start} > 1$ and $\mathbf{attack}_{end} = n$. In the interim FDI attack, the attacker ends the attack within the range of the vector. Here, $\mathbf{attack}_{start} >$ and $\mathbf{attack}_{end} < n$. A stealthy FDI attack is the most deceptive attack, where the attacker hides the attack from the defence mechanism. In a stealth attack, the attacker injects false data within the stability range. An attacker who has complete knowledge of the system injects data, such that the value is within the limits of stability, as shown in (3). Another method of stealth attack consists of two parts; the first is to perform the attack on the controller outputs, and the second is to stealthily hide the attack's impact from the controller.

$$\mathbf{A}_{min} < \mathbf{C}_i < \mathbf{A}_{max} \quad (3)$$

Unconstrained and constrained FDI attacks are the other types of FDI attacks. In unconstrained FDI attacks, the attacker has access to all the variables in the communication links, whereas in constrained FDI attacks, the attacker gains only limited access [23].

5.2. Man in the middle attack

In a MITM attack, an attacker aims to intercept and steal private information exchanged between two parties, as shown in Fig. 7. This involves two key steps: first, infiltrating the communication channel or intercepting data traffic, and second, decrypting the transmitted information. The attacker's motives can vary but often include stealing identities, altering login credentials, or gaining control of financial transactions. Attackers employ various methods to infiltrate networks,

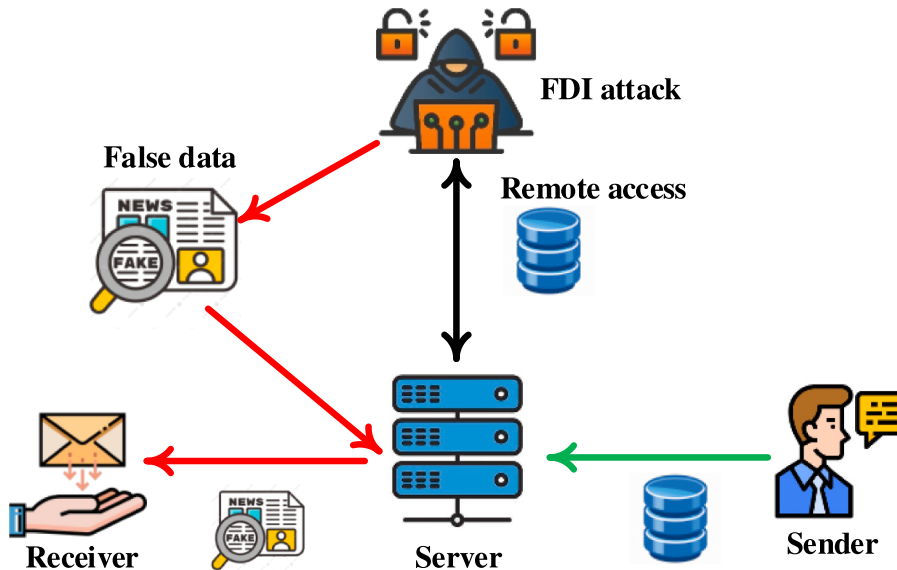


Fig. 6. Representation of an FDI attack.

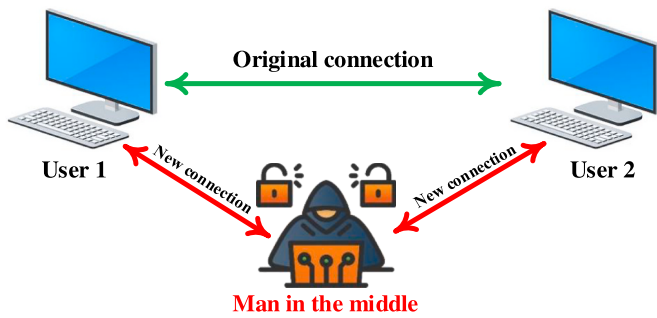


Fig. 7. Representation of a Man-in-the-Middle attack.

such as IP spoofing, Address Resolution Protocol (ARP) spoofing, and Domain Name System (DNS) spoofing. IP spoofing involves the attacker, positioned between the two communicating parties, impersonating both the sender and receiver by falsifying their IP addresses. This makes the communication appear legitimate to both parties. ARP, Address Resolution Protocol, is used to link a device's IP address to its media access control address, a crucial step in network communication [12].

When a device sends an ARP request, it receives an ARP response from the corresponding device, and these exchanges are stored in an ARP cache. An attacker gaining access to this cache can link their own media access control address to a device's IP address, effectively granting them access to data shared between parties. DNS spoofing is another technique where users are redirected to a fraudulent account controlled by the attacker due to manipulated domain names on the server. This allows the attacker to steal any confidential information the user enters, such as login credentials. Once inside the network, the attacker needs to decrypt the messages being transmitted. Several methods are used for this, including Hypertext Transfer Protocol Secure (HTTPS) spoofing, Secure Sockets Layer (SSL) hijacking, SSL stripping, and SSL beast [13].

5.3. DoS attack

DoS attack aims to disrupt a service's availability to legitimate users by overwhelming the server with bogus requests. This type of attack targets the "availability" aspect of information security [41]. While it doesn't compromise confidentiality or integrity, it does waste time and computational resources. Attackers often use DoS attacks to halt system performance, cause financial damage, or expose system vulnerabilities. A visual representation of a DoS attack, where an attacker uses multiple compromised devices to flood a server with requests, is shown in Fig. 8. DoS attacks can be categorized based on their target within the communication system. Application layer attacks target specific applications, often by flooding them with HTTP requests. Network layer attacks exploit weaknesses in communication protocols to disrupt the entire network. Volumetric attacks, the most common type, target the communication channel's bandwidth, creating network congestion. Various techniques, such as SYN scans, smart attacks, teardrop attacks, ARP attacks, and Fraggle attacks, can be used to execute DoS attacks. A distributed DoS attack is a related variant where multiple compromised devices are used to launch the attack, unlike a typical DoS attack which originates from a single source.

Attackers can destabilize CPS using DoS attacks in a couple of ways. One method involves saturating the communication channel with fake data, effectively blocking it. Another more subtle approach involves inserting a buffer into the communication flow, briefly halting communication between nodes and causing delays in information transfer. This delay can disrupt microgrid control operations and create instability. The buffer method is particularly dangerous due to its difficult detection. Various mitigation strategies have been proposed to counter DoS attacks on CPS, including methods like packet marking to trace packet origins and policy reinforcement learning that combines

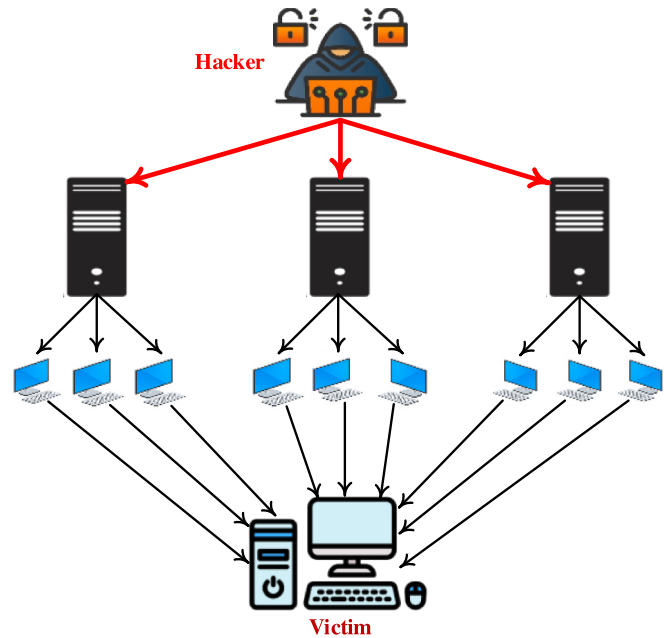


Fig. 8. Representation of a DoS attack.

event-triggering control, robust control theory, and game theory to detect these attacks by establishing a relationship between the cyber and physical components of the system.

5.4. The impact of attacks on IES performance

While the discussed attacks (FDI, DoS, and MITM) originate in the cyber layer, their consequences in an IES are uniquely devastating due to cross-carrier interdependencies. For instance, an FDI attack targeting the gas-flow sensors of a CHP unit does not only disrupt the gas sub-system but can trigger an immediate voltage instability in the electrical grid and a temperature drop in the district heating network. Similarly, a DoS attack that delays control signals in a multi-carrier EH can cause 'energy-imbalance cascades,' where the slower response time of thermal and gas actuators (minutes to hours) fails to compensate for the rapid fluctuations in electrical demand. These cross-domain impacts underscore that IES performance is not just about data integrity, but about maintaining the physical equilibrium across multiple energy vectors simultaneously.

5.5. Real-world attacks

Again, cyberattacks on energy infrastructure represent one of the most critical threats to the economic stability and national security of countries worldwide. Several high-profile incidents have demonstrated the severity of these threats [49,50]. This section provides an in-depth analysis of real-world cyberattacks on IESs, examining their impact and the strategies employed for mitigation and recovery.

As illustrated in Table 7, cyberattacks on energy systems have evolved in complexity and scope, posing significant risks to critical infrastructure. The 2015 cyberattack on Ukraine's power grid [51], the 2021 ransomware attack on Colonial Pipeline [52], and the 2022 cyber incidents targeting renewable energy systems exemplify the diverse and escalating nature of cyber threats in the energy sector [53]. While more recent incidents have occurred, such as the 2024 Ukrainian district heating attack, the 2025 cyber intrusion on Poland's power grid and renewable integration, the 2025 ICS breaches in Canada, and global ransomware attacks on utilities in 2025, detailed technical data for these events is often limited due to security and confidentiality concerns. These attacks not only disrupt essential energy supplies but also impose

Table 7

Real-world examples of cyber warfare.

Year	Country/ Region	Targeted Infrastructure	Attack Type/Method	Impact	Availability of Data	Notes/Significance
2015	Ukraine	Power Grid	Malware (BlackEnergy), ICS manipulation	Major blackout affecting ~225,000 customers	High	One of the first successful cyberattacks on a national power grid; well-documented, widely analyzed in literature
2021	United States	Major fuel pipeline (oil & refined products)	Ransomware attack (DarkSide) targeting IT systems, leading to shutdown	Temporary shutdown of pipeline operations, fuel shortages/panic buying; declared national emergency	High/ Moderate	Ransomware disrupted critical energy supply chain; forced shutdown and ransom payment ~\$5M
2022	Europe	Renewable energy generation systems	Cyber intrusion/attack on operational technology	Production disruptions in renewable energy output	Low/ Moderate	Demonstrates vulnerability of renewable energy systems to cyberattacks; limited technical data publicly available
2024	Ukraine	District Heating System	Malware targeting heating controls	Disruption of heating/water for ~600 buildings for 48+ hours	Moderate	Demonstrates attack on thermal energy infrastructure, not just electricity; recent example of ICS targeting
2025	Poland	Power Grid & Renewable Integration	Cyber intrusion targeting communication and control systems	No widespread blackout, but revealed vulnerabilities in grid operations	Low	Highlights evolving attack tactics on integrated energy systems; latest publicized incident
2025	Canada	Water & Energy ICS	Multiple ICS breaches including pump and sensor manipulation	Operational disruptions in critical facilities	Low	Shows trend of ICS attacks in multiple sectors; emphasizes global threat to critical infrastructure
2025	Global Utilities	Various Utilities	Ransomware and cyber intrusions	Temporary service disruption and threat exposure	Low	Illustrates increased frequency of attacks on utilities; supports need for AI-based adaptive defenses

far-reaching consequences on economic stability, national security, and societal well-being.

Indeed, the attack methodologies employed by cybercriminals vary significantly. For instance, BlackEnergy malware, responsible for the Ukraine power grid attack, directly infiltrated and manipulated power control systems, leading to widespread blackouts. Similarly, the DarkSide ransomware, which targeted the Colonial Pipeline, encrypted critical operational data, demanding ransom payments to restore functionality. More recent cyberattacks, such as distributed DoS assaults on renewable energy systems, highlight a strategic shift in cybercriminal activity towards modern, decentralized energy infrastructures, exploiting vulnerabilities inherent in emerging technologies. Despite the devastating impact of these incidents, response and recovery measures underscore the importance of proactive cybersecurity preparedness. Governments and energy organizations have adopted various mitigation strategies, including:

- Manual recovery operations to restore functionality when automated systems are compromised,
- Enhanced security protocols to strengthen system defences,
- Advanced threat detection mechanisms leveraging AI and ML for real-time threat identification, and
- High financial investments in cybersecurity resilience to bolster defence mechanisms against evolving threats.

However, as cyber threats become more sophisticated, ensuring the security of IECPSs requires continuous adaptation, investment in cutting-edge cybersecurity technologies, and international cooperation to develop cohesive regulatory frameworks and shared security standards. As such, Table 7 provides a detailed overview of some of the most significant cyberwarfare incidents affecting the global energy sector.

6. Analysis of operational safety in IECPS frameworks

Analyzing and evaluating the security of IECPSs encompassing both reliability and vulnerability assessments is essential for the development of systems that are both robust and resilient. This section provides a comprehensive examination of these aspects, emphasizing their critical role in ensuring secure and dependable IECPS operation.

6.1. IECPS reliability assessment

Reliability analysis uses qualitative or quantitative indicators to assess energy supply interruption risks, guiding IES planning and operation. Unlike single energy system reliability analysis, IES analysis must consider differing energy properties, time scales, evaluation indicators, and complex energy coupling. IECPS reliability analysis, from a cyber-physical perspective, must also incorporate cyber system reliability and cyber-physical interaction, increasing complexity. Numerous studies address IES reliability, with [54] summarizing challenges in modelling, algorithms, and indices, offering future research recommendations. Reliability modelling methods for IES encompass a range of analytical approaches, including Markov processes [55,56], graph theory [57], and reliability theory [33,58]. In Ref. [55], the authors employed a sequential Monte Carlo simulation to evaluate the reliability of electrical IES, introducing an optimal load-shedding index as a performance metric. Meanwhile, the study in Ref. [56] developed a stochastic network capacity model for gas pipeline systems by integrating graph-theoretical concepts with Markov-based techniques, resulting in the formulation of comprehensive reliability indices to assess system performance under uncertainty. References [59–61] employed multi-state system models to evaluate the reliability of IES, extending these models to account for multi-energy conversion processes. In the context of CPSs, reliability assessment incorporates both physical system failures [62] and network-related disruptions, including communication delays, interruptions, and cyberattacks [63–65]. A reliability framework for CPES was proposed that integrates physical component failures with load reconfiguration attacks, with [62] being among the first to examine electrical system reliability under such malicious conditions. In Ref. [66], the authors introduced a variable sensitivity index and a branch correlation matrix to identify and secure critical information transmission paths, thereby enhancing electrical system resilience. Research on IECPS reliability assessment, however, remains relatively underdeveloped. In Ref. [67], the authors reviewed reliability evaluation techniques within the context of the energy internet, presenting key insights and conceptual frameworks for IECPS reliability under cyber-physical integration specifically considering data and business process flows. Furthermore [68], investigated IECPS reliability through the lens of intelligent communication, proposing the reconstruction of communication architectures using intelligent agents. While extensive

research exists for IES and CPES reliability, the evaluation of IECPS reliability in cyber-physical environments is still in its nascent stage, despite its critical importance in practical engineering applications.

6.2. IECPS vulnerability assessment

Vulnerability is defined as a system's susceptibility to failures, threats, disasters, and attacks, or the degree of low reliability [69], essentially measures a system's weak links. In IECPS, coupling components like CHP units act as both energy flow sources and critical information nodes, significantly increasing vulnerability. Current IES vulnerability analysis primarily uses energy hub models and complex network theory [70]. CPES vulnerability assessment is more extensive, employing complex network theory [71], ML [72], simulation, and

optimization [73]. Reference [74] summarized assessment methods and indicators for power information physical system vulnerability. Authors in Ref. [71] used complex network theory to identify key power grid nodes, establishing a cascade fault model based on AC power flow and network topology weighted admittance. Reference [73] proposed a framework for identifying vulnerable electrical system components, considering component interaction and cascading failures, using game theory-based optimization. Reference [75] proposed a vulnerability assessment framework for wide-area power systems, analysing the impact of communication network delay and interruption on reliability.

6.3. Summary of research on security performance

The preceding analysis of IES, CPES, and IECPS security performance

Table 8
Overview of research on IECPS's cyberattack and defence.

Ref.	Sub-energy System	Attack Target	Attack Method	Defense Strategy	Analytical Method
[29]	Thermal	Dispatch/operation info, Communication/physical line	Confidentiality, Internality, Availability	Innovative detection	Intelligent Algorithms
[44]	Gas, Electrical, Thermal	Supply/Demand Info, Dispatch/operation info, Communication/physical line	Confidentiality, Internality, Availability	Resource Allocation, Innovative detection, Coordinated operations	Simulation, Intelligent Algorithms
[76]	Electrical, Thermal	Dispatch/operation info, Communication/physical line	—	Innovative detection, Coordinated operations	Intelligent Algorithms
[77]	Gas, Electrical	Dispatch/operation info	Confidentiality, Internality	Innovative detection	Simulation
[78]	Electrical, Cooling	Dispatch/operation info	Confidentiality, Internality, Availability	—	Intelligent Algorithms
[79]	Gas, Electrical	Supply/Demand Info	Confidentiality, Internality	Resource Allocation	Game Theory
[80]	Gas, Electrical	Communication/physical line	Availability	Resource Allocation	Game Theory
[81]	Gas, Electrical	Supply/Demand Info	Confidentiality, Internality	Resource Allocation	Game Theory
[82]	Electrical, Thermal, Cooling	Supply/Demand Info	Confidentiality, Internality	Resource Allocation, Coordinated operations	Simulation
[83]	Electrical, Cooling	Dispatch/operation info, Communication/physical line	Confidentiality, Internality, Availability	Innovative detection	Game Theory, Simulation, Intelligent Algorithms
[84]	Gas, Electrical, Thermal,	Dispatch/operation info	Confidentiality, Internality	Resource Allocation, Resource Allocation	Game Theory
[85]	Gas, Electrical, Cooling	Communication/physical line	Confidentiality, Internality	Resource Allocation, Coordinated operations	Game Theory
[86]	Gas, Electrical,	Dispatch/operation info	Confidentiality, Internality,	Resource Allocation, Innovative Detection	Game Theory, Intelligent Algorithms
[87]	Gas, Electrical, Cooling	Supply/Demand Info, Dispatch/operation info,	Confidentiality, Internality	Resource Allocation	Intelligent Algorithms
[88]	Electrical, Thermal	Dispatch/operation info	Confidentiality, Internality, Availability	Coordinated operations	Game Theory, Intelligent Algorithms
[89]	Electrical,	Dispatch/operation info, Communication/physical line	Confidentiality, Internality, Availability	Innovative detection	Simulation, Intelligent Algorithms
[90]	Gas, Electrical, Hydrogen	Dispatch/operation info	Confidentiality, Internality, Availability	Resource Allocation, Coordinated operations	Simulation, Intelligent Algorithms
[91]	Gas, Electrical, Thermal, Cooling, Hydrogen	Supply/Demand Info, Dispatch/operation info, Communication/physical line	Confidentiality, Internality, Availability	Resource Allocation, Coordinated operations	Game Theory, Intelligent Algorithms
[92]	Gas, Electrical, Thermal	Supply/Demand Info, Dispatch/operation info, Communication/physical line	Confidentiality, Internality, Availability	Innovative detection, Coordinated operations	Simulation, Intelligent Algorithms
[93]	Gas, Electrical	Supply/Demand Info, Dispatch/operation info, Communication/physical line	Confidentiality, Internality, Availability	Coordinated operations	Game Theory, Intelligent Algorithms
[94]	Gas, Electrical	Supply/Demand Info, Dispatch/operation info,	Confidentiality, Internality, Availability	Resource Allocation, Coordinated operations	Game Theory, Intelligent Algorithms
[95]	Electrical, Thermal (Building HVAC)	Sensor data, Control signals, Energy performance data	Confidentiality, Integrity, Availability	Physics-informed attack detection based on energy performance evaluation	Physics-informed Dynamic Bayesian Network (PIDBN)
[96]	Electrical, Thermal (Smart Buildings)	Control signals, System states, Dispatch/operation info	Integrity, Availability	Cyber-resilient control using adaptive predictive control	Adaptive Model Predictive Control (AMPC)
[51]	Gas, Electrical, Thermal (IE-CPS)	Communication network, Dispatch/operation info, Supply/Demand data	Confidentiality, Integrity, Availability	Coordinated defense mechanisms, Resilient operation strategies	Review, Simulation, Intelligent Algorithms
[97]	Electrical, Thermal (Building Energy Systems)	Physical components, Sensors, Communication links	Integrity, Availability	Real-time attack and fault detection using hardware-in-the-loop	Hardware-in-the-Loop Experiments, Simulation
[98]	Electrical, Renewable Energy Systems	Control systems, Communication infrastructure	Confidentiality, Integrity, Availability	Cyber risk mitigation and security enhancement strategies	Risk Analysis, Survey-based Analysis

reveals that IECPS research remains preliminary despite progress in IES and CPES. IECPS security performance analysis largely relies on and extends existing methods like energy-hub and complex network analysis from IES and CPES, with the extended energy-hub model analysing information-physical coupling and CPES complex network analysis adapted for IES multi-energy coupling. However, current IECPS security performance analysis using these approaches is insufficient due to functional fragmentation in both framework and model. For example, the energy hub's cyber-physical coupling description is inadequate, and complex network depictions lack complete multi-energy flow and cyber-physical coupling representation. Future research must refine security performance analysis models and frameworks to simultaneously reflect both cyber-physical and multi-energy flow coupling in IECPS. Furthermore, a comprehensive reliability/vulnerability evaluation index that fully captures these couplings is currently lacking, representing a key future research priority.

7. Cyberattack mitigation and security architecture in IECPS

IECPSs are exposed to three principal categories of security threats: natural disasters, random equipment failures, and malicious cyberattacks with the latter constituting the most direct and severe risk. In the context of electrical systems, cyberattacks typically involve unauthorized intrusions into communication and control infrastructures, aiming to disrupt the functionality of CPES by exploiting vulnerabilities within information and communication networks [45]. These attacks threaten the core cybersecurity principles of confidentiality, integrity, and availability. Common cyberattack types in CPES include DoS attacks, MITM attacks, and FDI attacks. Although attacks on both IECPS and CPES compromise the same security triad, IECPS-specific threats are more complex due to the intricacies of multi-energy coupling, which links electricity, gas, and heat subsystems.

Table 8 presents a comprehensive overview of recent research on cyberattacks and defense mechanisms in IECPSs, covering a wide range of sub-energy systems and analytical approaches. As shown in Table 8, most studies focus on attacks targeting dispatch and operational

information, supply-demand data, and communication or physical links, particularly within electrical, gas, and thermal subsystems [76–97].

From the attack perspective, confidentiality, integrity, and availability (CIA) constitute the dominant threat dimensions across IECPSs. Integrity and availability attacks are emphasized in many recent works due to their direct impact on system stability and real-time operation [70,72,78,80,87,90]. These findings highlight the increasing vulnerability of sensor measurements, control signals, and communication infrastructures in integrated energy environments.

Regarding defense strategies, innovative detection mechanisms are widely adopted to identify cyber intrusions and abnormal system behaviours [27,42,68,69,75,81,87,90]. In parallel, resource allocation and coordinated operation strategies are frequently employed to enhance system resilience and mitigate cascading effects under cyberattack scenarios [42,71–74,76,82,83,85,86,88,89]. These coordinated approaches are particularly effective in multi-energy systems where interdependencies among subsystems amplify cyber risks.

In terms of analytical methods, intelligent algorithms and simulation-based techniques dominate the literature [27,42,68,70,75,79,80,82,84]. Game-theoretic models are extensively used to analyze strategic interactions between attackers and defenders, especially in resource allocation and security investment problems [71–73,76,78,85,86]. More recent studies introduce advanced hybrid frameworks, such as physics-informed dynamic Bayesian networks [95], adaptive model predictive control [96], and real-time hardware-in-the-loop experiments [45], which significantly improve the realism and robustness of cyber defense evaluations. Additionally, comprehensive review and risk analysis studies provide valuable insights into emerging cyber threats and defense trends in IECPSs [97,98]. A visual summary of this data is presented in Fig. 9.

7.1. IECPS attack method

IECPS cyberattacks, categorized according to cybersecurity's three elements [45], include confidentiality attacks (unauthorized

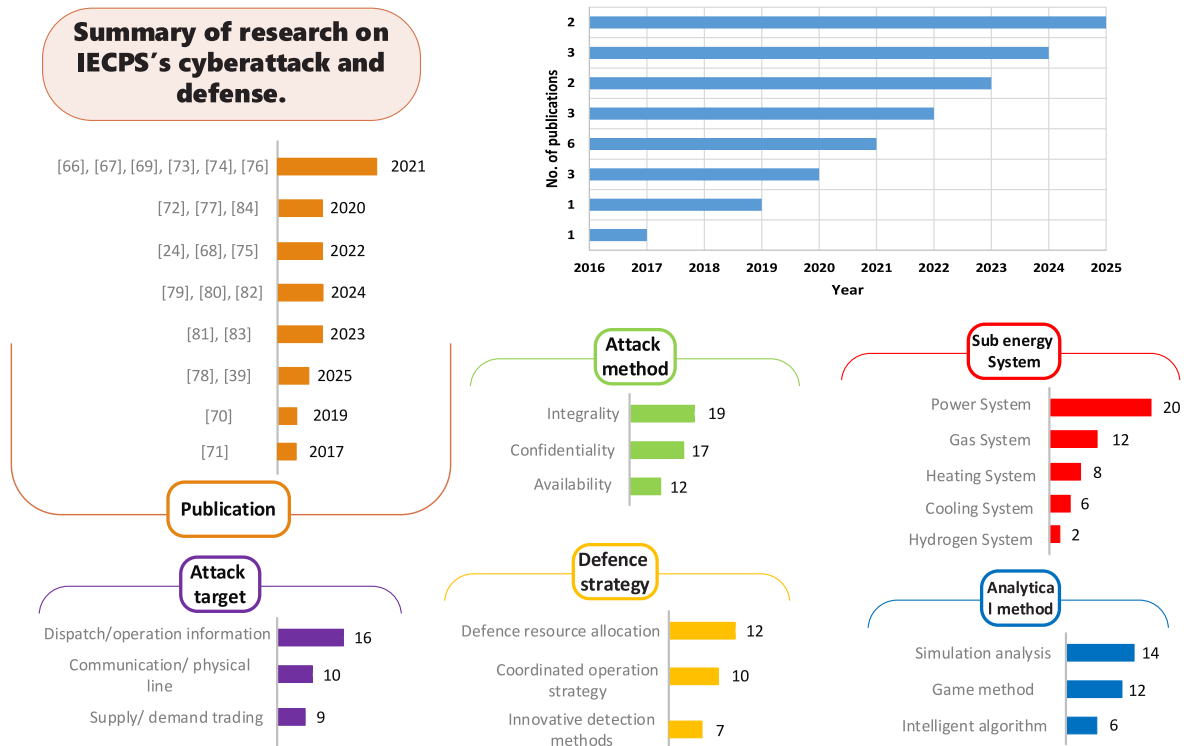


Fig. 9. A visual summary of the data presented in Table 5.

information access, often combined with other attacks), integrity attacks (information tampering, like FDI, and availability attacks (preventing timely data/signal access, including DoS and delay attacks). Currently, FDI and DoS attacks are the main methods against IECPS. Unlike electrical systems or other sub-energy system cyberattacks, IECPS attacks differ in target. As detailed in Table 5, existing research on cyberattacks in IECPS identifies three primary target layers: 1) Market and transactional information, including inter-subsystem supply and demand data, energy consumption and pricing details, contractual information, and data related to coordinated operations in large-scale IECPS. Attacks on this layer are generally classified as market or transactional cyberattacks; 2) Dispatch and operational control information within IES, where adversaries seek to disrupt system scheduling, control commands, or optimization procedures; and 3) Communication infrastructure and physical transmission lines, where attackers exploit vulnerabilities in the cyber-physical interface, targeting both digital communication networks and physical energy delivery components. Attacks targeting inter-subsystem supply, demand, and transactional information primarily compromise the market and transaction layer of IECPS. For instance, the study in Ref. [99] proposed defence mechanisms to safeguard energy contract and supply data within EG-IES, while [75] demonstrated a FDI attack on gas system demand and scheduling information. This type of attack disrupts supply-demand balance and enables price manipulation within the market. Cyberattacks on IES dispatch and operational data directly interfere with system control and performance. As illustrated in Ref. [77], FDI attacks can manipulate input/output signals of electric-to-gas and gas-to-electric conversion scheduling equipment. These attacks target critical operational data such as electrical and gas demand, energy output, and pricing transmitted to and from the dispatch centre. Finally, attacks on IECPS communication infrastructure and physical lines undermine the system at the device and communication level. For example, the authors in Ref. [78] presented a DoS attack on building energy control systems, along with physical sabotage that induces interlocking failures in EG-IES, further emphasizing the vulnerability of the cyber-physical interface.

7.2. Detection and defence strategy

Table 9 categorizes the cyberattack detection and defence approaches for IECPSs into three main strategies: optimization of defence resource allocation, implementation of intelligent detection techniques, and enhancement of collaborative scheduling mechanisms. These categories reflect a comprehensive framework aimed at strengthening system resilience. Optimizing the deployment of defence resources ensures efficient use of protective assets across the system. Intelligent detection techniques leverage advanced analytics and ML to identify potential

threats in real time. Meanwhile, improving collaborative scheduling enhances coordination among subsystems, enabling adaptive responses to cyber threats and maintaining stable operations under adverse conditions. Reference [76] proposed a three-layer defender-attacker-operator optimization for budget allocation to protect critical fuel supply information. Intelligent algorithms are suitable for IECPS cyberattack detection in complex environments. The authors in Ref. [77] proposed two EG-IES FDI defence strategies: supervised ML using convolutional neural networks and wavelet transform, and a hybrid neural network using unsupervised learning. The study in Ref. [81] used ML to detect building-level IES cyberattacks. For DoS and replay attacks on building-level IES energy management systems [83], used Boolean identification for electrical faults and attacks, followed by ML for attack method identification. Regarding collaborative scheduling [76], proposed a fully distributed economic scheduling strategy for IES to address non-collusive and collusive attacks, including a privacy protection protocol and a neighbourhood observation-based detection method for cyberattack-induced bad behaviour. Reference [80] proposed protecting vulnerable gas-power infrastructure components and deploying corrective measures to resist IECPS attacks.

Current IECPS cyberattack and defense research, primarily focused on EG-IES, often uses optimization or game theory for attack resource allocation and rarely targets deeper cyber system levels like control methods and logic. While secure resource allocation and collaborative defense strategies are readily implementable due to their maturity in CPES and other industrial control systems, and the latter leverages multi-energy coupling and existing IES research, further research on cyber and control level vulnerabilities and security is necessary due to cyberattacks directly targeting system control and communication. Specifically, future research should address: (a) the diverse communication protocols (DNP3, IEC 61850, IEC 60870 [100,101] for power, Modbus [102–104] (for thermal) and media (dedicated fibre/internal network for power, Internet for thermal with encryption before entering the IES internal network) in IES, while integration enhances security, it doesn't guarantee immunity [105–108] due to multi-protocol/process information interaction increasing vulnerability; (b) backward compatibility requirements in electrical (and other energy) systems leading to less secure designs [109], and lagging information and communication technology in thermal systems making IES's cyber layer more vulnerable due to the "barrel principle"; and (c) the lack of systematic communication design and targeted research on cross-system information interaction between thermal and electrical systems in IES, intensifying cybersecurity vulnerability.

As shown in Table 9 Firewalls, while fundamental, are limited against sophisticated attacks. Intrusion Detection/Prevention Systems (IDS/IPS) offer continuous monitoring but require careful configuration to avoid false positives. Encryption is crucial for data protection,

Table 9

A Comparative analysis of cybersecurity defense techniques in multi-energy carrier systems.

Defence Technique	Working Mechanism	Key Advantages	Critical Limitations/Trade-offs	Efficiency Level & Context
Firewalls	Packet filtering and network traffic control.	Low cost; easy to deploy for basic perimeter security.	Vulnerable to insider threats and sophisticated FDI attacks.	Moderate; foundational but insufficient for OT safety.
IDS and IPS	Monitoring network traffic for signature-based or anomalous patterns.	Continuous monitoring; effective against known exploits.	High rate of false alarms; requires complex tuning for energy systems.	Good; requires integration with physics-based residuals.
Encryption	Cryptographic protection of data during transit and storage.	Prevents eavesdropping and MITM attacks.	Adds communication latency; does not prevent data manipulation at the source.	High; essential for data integrity across energy hubs.
AI/ML-Based Detection	Pattern recognition to identify zero-day threats and anomalies.	Adaptive to evolving threats; high accuracy in complex data.	High computational overhead; requires massive datasets and periodic retraining.	Excellent; but limited by real-time processing constraints.
MFA (Multi-Factor Auth)	Verification through multiple independent credentials.	Strong defense against credential theft and unauthorized access.	May cause operational friction and delay emergency response for operators.	High; recommended for administrative cyber-access.
Cyber Hardening	Reducing attack surfaces by patching and removing vulnerabilities.	Proactively minimizes system weaknesses and entry points.	Costly; requires frequent maintenance and system downtime for updates.	High; a core preventive measure for long-term resilience.
Redundancy & Recovery	Using backup assets and self-healing control strategies.	Ensures continuity of service and minimizes physical impact.	Reactive nature; does not stop the attack but mitigates its results.	Critical; the last line of defense for physical system safety.

particularly against man-in-the-middle attacks, but doesn't address insider threats. ML and AI-based detection offer adaptability and accuracy but demand significant resources. Multi-Factor Authentication (MFA) strengthens access control but can impact usability. Cyber Hardening minimizes vulnerabilities but requires ongoing updates. Finally, Redundancy and Recovery Strategies are essential for mitigating the impact of successful attacks, although they are reactive rather than preventive.

As analyzed in Table 6, the transition from traditional IT security to integrated OT resilience requires a careful evaluation of methodological trade-offs. While conventional tools like Firewalls and Encryption provide necessary perimeter defense, they often fail to address the dynamic coupling between cyber commands and physical energy flows in IECPS. For instance, the deployment of AI-based detection while superior in accuracy must be weighed against its computational latency, which can be critical in real-time energy balancing. Furthermore, a shift from purely preventive measures (like Hardening) to resilient-reactive strategies (like Redundancy) is observed, highlighting the field's progression toward maintaining physical stability even under successful cyber breaches.

8. Cybersecurity-oriented resilience and recovery frameworks

The coordinated operation and recovery of IECPSs in the face of cyberattacks constitutes a critical and evolving area of research. Unlike single-energy systems, IECPS exhibit increased structural complexity and reduced operational flexibility, thereby requiring more advanced and integrated strategies to ensure system resilience and secure recovery. Resiliency, defined differently across research fields, refers to energy systems maintaining full or partial operation under abnormal conditions (physical faults, natural disasters, and cyberattacks). Reference [81] defined CPES resiliency as maintaining core functions under attack and timely restoring regular operation. Despite variations, resiliency definitions emphasize the inevitability of successful cyberattacks and the system's recovery ability. Thus, IECPS resiliency under cybersecurity threats is defined as the ability to recover from cyberattacks. In electrical systems, authors in Ref. [88] used adaptive fuzzy neural networks to reconfigure emergency power supply, electric vehicles, and energy storage for improved resiliency. IECPS resiliency analysis, unlike CPES, must consider coupling equipment vulnerability and inter-system information barriers.

Considering renewable energy uncertainty, the authors in Ref. [85] proposed a two-stage distributed robust method for IES resiliency under cyberattack. For building-level IES FDI attacks, the work in Ref. [82] proposed resiliency by limiting actual vs. predicted typical load deviation. However, these methods ignore EG-IES's complex energy/information flow coupling. IECPS recovery strategies must consider multi-energy system coupling and cooperation. Table 10 highlights the critical role of advanced mechanisms in bolstering cybersecurity defenses, particularly within complex systems like IECPS. Smart Redundancy ensures data integrity in the face of attacks like FDI, while

Table 10
Advanced mechanisms for enhancing cybersecurity in IECPS.

Mechanism	Description	Cybersecurity Applications
Smart Redundancy	Replacing suspicious data with trusted data from backup sensors.	Recovery from FDI attacks.
Automated Network Reconfiguration	Redirecting traffic away from affected devices.	Mitigating the impact of DoS attacks.
Predictive Analytics	Predicting future threats based on historical attack data.	Preventing attacks before they occur.
Self-Healing Systems	Detecting failures and autonomously restarting affected systems.	Reducing downtime after an attack.

Automated Network Reconfiguration provides a swift response to DoS attacks by redirecting traffic away from compromised nodes [110]. Predictive Analytics takes a proactive stance, leveraging historical data to anticipate and thwart future threats before they materialize [111]. Finally, Self-Healing Systems minimize downtime and disruption by autonomously detecting failures and restoring affected systems [112].

9. Case study: The 2015 Ukraine power grid cyberattack

The 2015 Ukraine power grid cyberattack was a watershed moment in the history of cyber warfare [113]. It was the first publicly acknowledged successful cyberattack to cause a widespread power outage, impacting approximately 230,000 people in Western Ukraine. This attack, attributed to the Russian hacking group Sandworm, demonstrated the vulnerability of critical infrastructure to sophisticated cyberattacks and highlighted the potential for significant disruption to essential services.

9.1. Attack history

The attackers employed a multi-stage approach, leveraging spear-phishing emails to gain initial access to the IT networks of three regional power distribution companies. Once inside, they moved laterally, gaining control of SCADA systems responsible for managing the grid's operations. They then launched a coordinated attack, disrupting the power supply, manipulating control systems, and even rendering the SCADA systems inoperable by overwriting firmware. This attack combined conventional malware with destructive techniques, showcasing a high level of planning and coordination. This attack unfolded in a series of carefully orchestrated steps:

A. Infiltration:

The attackers initiated the operation by sending targeted phishing emails to employees of Ukrainian power distribution companies. These emails contained malicious attachments laced with the BlackEnergy malware. Unsuspecting employees opened the attachments, unknowingly granting the attackers a foothold within the companies' IT networks.

B. Escalation and Control:

Once inside, the attackers exploited vulnerabilities to steal login credentials, allowing them to move deeper into the networks and gain access to critical systems, including the SCADA systems responsible for controlling the power grid. The attackers then proceeded to disable the SCADA systems, effectively taking remote control of the power grid operations. This allowed them to manually manipulate circuit breakers and disconnect substations, causing widespread power outages.

C. Disruption and Delay:

To hinder recovery efforts, the attackers deployed the KillDisk malware, which wiped critical data and operating systems on several servers, making it difficult for engineers to restore power quickly [113]. Simultaneously, a DoS attack was launched against the power companies' call centres, flooding them with traffic and preventing customers from reporting the outages, further delaying response and recovery efforts.

This multi-faceted attack demonstrated a high level of planning and coordination, combining conventional malware with destructive techniques to achieve maximum impact and disruption.

9.2. Impact of the attack

The most immediate impact was the disruption of electricity supply to approximately 230,000 people across Western Ukraine. Homes, businesses, and critical services were left without power for several hours, causing significant inconvenience and disruption to daily life. The attackers' sabotage of SCADA systems resulted in the loss of remote-

control capabilities for the affected power distribution companies. This meant that engineers could no longer monitor and manage the grid remotely, hindering their ability to respond effectively to the outages. With remote control disabled, emergency response teams were forced to resort to manual operations to restore power. This involved physically traveling to substations and manually resetting equipment, a time-consuming and laborious process that significantly delayed the restoration of services.

This attack served as a stark reminder of the vulnerability of critical infrastructure to cyberattacks and the potential for widespread disruption to essential services. It underscored the urgent need for enhanced cybersecurity measures to protect these vital systems and ensure the continuity of essential services.

9.3. Attack mitigation/prevention with modern methods

The 2015 Ukraine power grid cyberattack was a wake-up call, demonstrating the vulnerability of critical infrastructure to sophisticated cyberattacks. However, modern cybersecurity methods and technologies offer effective ways to prevent or mitigate such attacks.

A. Preventing the Attack with AI and ML:

ML models can analyze network traffic patterns and system behaviour to identify anomalies that deviate from established baselines. This allows for the detection of malicious activities, such as malware infections or unauthorized access attempts, even before they fully manifest.

AI-powered email filtering systems can analyze email content, sender reputation, and other factors to identify and block phishing emails with a high degree of accuracy, preventing them from reaching employees' inboxes.

Even if attackers manage to steal login credentials, implementing MFA would have thwarted their attempts to access critical systems. MFA requires users to provide additional verification factors, such as a one-time code or biometric authentication, making it significantly harder for attackers to gain unauthorized access.

B. Reducing the Attack's Impact Through Improved Security Strategies:

Isolating the Industrial Control System network from the administrative IT network would have created a barrier, preventing the attackers from easily moving laterally from the compromised IT network to the critical SCADA systems. This segmentation limits the impact of an attack by containing it within a specific network segment [114].

Deploying robust IDS/IPS solutions, such as Snort or Suricata, would have enabled real-time monitoring of network traffic for malicious activity [112]. These systems can detect and block known attack signatures, as well as identify anomalous behaviour that could indicate an attack in progress.

C. Rapid Recovery Using Self-Healing Systems:

AI-powered self-healing systems can automate the recovery process by restarting compromised systems, restoring data from backups, and even proactively patching vulnerabilities. This minimizes downtime and reduces the impact of an attack. Big data analytics can be used to identify patterns and predict which system components are most likely to be targeted in an attack. This allows for proactive reinforcement of those components, making them more resilient to attacks.

Regular cybersecurity training and simulations can prepare response teams to handle cyberattacks effectively. By practicing incident response procedures in simulated attack scenarios, teams can develop the skills and knowledge needed to react quickly and efficiently, minimizing damage and downtime.

The 2015 Ukraine power grid cyberattack served as a stark warning,

exposing the vulnerability of critical energy infrastructure to sophisticated cyber threats. However, this incident also highlighted the potential of modern cybersecurity technologies and strategies to prevent or mitigate such attacks. By leveraging the power of AI and ML for proactive threat detection, implementing robust access controls like multi-factor authentication, deploying self-healing technologies for rapid recovery, and investing in comprehensive cybersecurity training and simulations, we can significantly enhance the resilience of our critical infrastructure and minimize the impact of future cyberattacks.

10. Future challenges

With the rapid advancement of digital technologies, the cybersecurity of IECPSs faces increasing challenges that require innovative solutions.

- **Enhanced Cybersecurity with Advanced AI:** With the growing sophistication of cyberattacks, developing advanced AI models using Deep Learning and Recurrent Neural Networks is essential to analyze operational data and predict potential threats. To address these challenges, standardized AI frameworks can be developed and trained on both real and synthetic data, integrating unsupervised and semi-supervised learning techniques to reduce reliance on labeled datasets. Additionally, AI-driven response modules can be implemented to autonomously take defensive actions upon detecting anomalies, enhancing the overall resilience and security of IECPSs.
- **Quantum Computing-Powered Cybersecurity:** With development of quantum computing, traditional encryption techniques such as Rivest-Shamir-Adleman and Advanced Encryption Standard will become vulnerable to cracking within a short period. Quantum Cryptography, such as quantum key distribution can be applied to ensure data protection against advanced cyberattacks. Quantum algorithms can improve big data analysis to detect suspicious activities in real time. Quantum AI techniques can be applied to create advanced detection systems capable of predicting and handling attacks with higher efficiency [115].
- **Cybersecurity Integration with Blockchain:** Blockchain technology can be used to create tamper-proof records that record all operational processes in integrated systems, preventing data manipulation by attackers [116]. Implementing Smart Contracts to ensure the secure execution of critical operations without the need for a central intermediary. Blockchain can enhance cyberattack detection by creating a reliable data distribution network that allows for the exchange of threat information between different systems. Implementing Decentralized Identity Systems to protect user data and energy infrastructure.
- **Developing Resilient and Adaptive Systems to Future Attacks:** Developing adaptive defense systems that adapt to new threats by automatically updating security databases. Applying Real-time Risk Assessment to evaluate the threat level and respond appropriately without delay. Establishing cooperation platforms between countries and companies to exchange cyberattack data and enhance security via collaborative intelligence. Developing global cybersecurity fusion centres to monitor and counter-attacks through integrated AI systems.
- **Physics-informed Intrusion Detection Systems (IDS):** A critical future direction for IECPS security is the development of physics-informed Intrusion Detection Systems (IDS). Unlike traditional IT-based IDS, these hybrid frameworks fuse cyber-layer features (e.g., protocol timing, authentication patterns) with physical residuals derived from multi-energy dynamics such as power flow, thermal gradients, and gas pressure. By utilizing grey-box digital twins and physical invariants, these systems can detect stealthy attacks that bypass cyber-defenses. Future research should prioritize reporting both detection quality (e.g., F1-score, latency) and physical impact

metrics (e.g., energy imbalance, pressure deviations) to provide a holistic view of system resilience.

- **Hardware-in-the-Loop (HIL):** presents a critical challenge in transitioning from pure software simulations to testing against physical constraints. To address this, standardized HIL benchmarks can be established to enable cross-laboratory evaluation of proposed systems. Controller-in-the-loop adversarial scenarios can be tested to simulate advanced attacks and assess system resilience. Additionally, integrating cyber-physical co-simulation allows near-real performance assessment, ensuring that the security frameworks are both scalable and deployable in real-world energy environments.
- Current regulatory frameworks exhibit significant gaps concerning integrated energy assets. While NERC CIP is robust for bulk power, it lacks coverage for behind-the-meter assets and thermal/hydrogen interfaces. Future governance must establish compliance roadmaps for Aggregators who manage multi-carrier loads, ensuring that data privacy for prosumers is balanced with the need for system-wide visibility.

11. Conclusions

The findings of this review reveal several critical gaps in the literature on cybersecurity for Integrated Energy Cyber-Physical Systems (IECPSs). Although existing studies have explored cybersecurity within cyber-physical system frameworks, they often remain generic and overlook the unique challenges posed by IECPS, including the lack of comprehensive models that capture complex multi-energy interactions, insufficient consideration of dynamic, real-time energy flows and control mechanisms, and limited cyberattack simulations that focus on basic information exchange while neglecting secondary equipment and control-level countermeasures. The non-standard physical structure and multi-energy coupling further increase the complexity of modeling, security assessment, and attack/defense analysis, requiring more sophisticated approaches. Future research should therefore focus on systematic cross-system communication design, incorporating multi-timescale operational and control dynamics, extending analyses to dynamic control levels with appropriate detection and defense strategies, and leveraging multi-energy coupling to enable cooperative scheduling, resilient control, and coordinated recovery, ultimately advancing IECPS cybersecurity through integrated multi-energy modeling, dynamic control analysis, and architecture-level system design.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Vladimír Bures reports financial support was provided by Technology Agency of the Czech Republic. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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